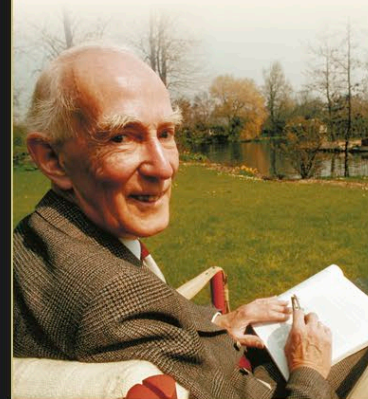


THE OORT CLOUD

Surrounding the solar system beyond the Kuiper Belt lies an enormous cloud of long-period comets known as the Oort Cloud. Its outer reaches extend to almost a light-year from the Sun, and it is thought to contain trillions of objects with a total mass of roughly five Earths. The Oort Cloud is impossible to observe directly, although there is strong evidence for it from the orbits of comets that pass through the inner solar system. Its existence was first suggested by Estonian astronomer Ernst Öpik in 1932, but it was also proposed independently by Jan Oort (see panel, right) in 1950. The comets in the Oort Cloud are thought to have originated much closer to the Sun, in the region where the giant planets now orbit. However, early in the solar system's history, as the giant planets migrated toward their current positions, close encounters with these planets pushed enormous numbers of the comets into highly elliptical orbits. In the outer solar system, these comets were only weakly bound by the Sun's gravity, so tidal forces from other stars and the Milky Way itself were able to act on them, gradually "circularizing" their orbits. Today, similar tidal effects occasionally knock comets out of the Oort Cloud toward the Sun. However, according to other theories, some Oort Cloud comets might have begun their lives in orbit around other stars and were later captured by the Sun's gravity.

JAN HENDRIK OORT

Jan Oort (1900–1992) was born in Franeker, the Netherlands, and spent most of his career at the University of Leiden, also in the Netherlands. Oort is mostly remembered for his idea that the solar system is surrounded by the vast symmetrical cloud of comets that was named after him. He is also famous as a pioneer of radio astronomy, for discovering the rotation of the Milky Way and estimating its distance and the direction of its center from Earth, and for discovering evidence that the universe contains "missing mass" (now known as dark matter).



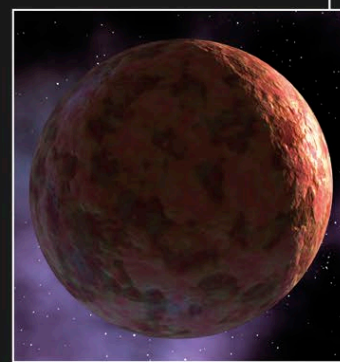
LONG-PERIOD COMET

Long-period comets, such as Hyakutake (left), typically approach the inner solar system from all directions and at high speeds, indicating that they come from a spherical region that surrounds the Sun at a vast distance—the Oort Cloud.



SEDNA

In 2003, astronomers searching for objects in the region of space beyond Neptune discovered a world about 8.4 billion miles (13.5 billion km) from the Sun but moving in an eccentric orbit that takes it out to a maximum distance of 87.1 billion miles (140.2 billion km). This was the most distant object yet found in the solar system. With an estimated surface temperature of -436°F (-260°C), it was also the coldest solar system body, so it was named after the Inuit goddess of the Arctic Ocean, Sedna. Some astronomers believe that Sedna could offer our first glimpse of an object from the inner Oort Cloud. However, its orbit is unusual even for an object from the innermost part of the Cloud, suggesting that it must have been disrupted in the past. Sedna's orbit is too remote for it to have been influenced by Neptune, but other possible explanations include disruption by the gravity of other stars or even by the influence of a large, as-yet-undiscovered planet far beyond Neptune.



RED BODY

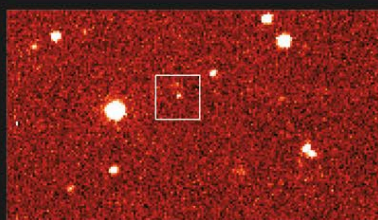
Sedna's diameter is estimated at 750–1,000 miles (1,200–1,600 km), and it has a dark red surface, as shown in this artist's impression.

comet's orbit takes it to the edge of the Oort Cloud

comet orbiting close to the plane of the solar system

inner cloud

outer cloud



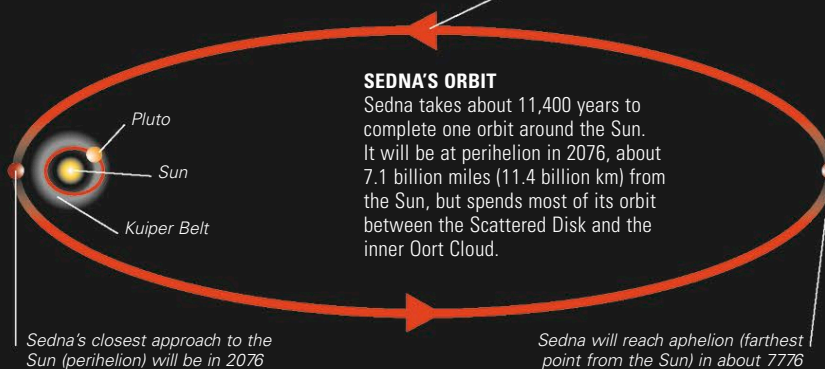
DISTANT OBJECT

When discovered on November 14, 2003, Sedna was nearly 90 times farther from the Sun than Earth, making it the most distant solar system object then observed.

Sedna takes approximately 11,400 Earth years to orbit the Sun

SEDNA'S ORBIT

Sedna takes about 11,400 years to complete one orbit around the Sun. It will be at perihelion in 2076, about 7.1 billion miles (11.4 billion km) from the Sun, but spends most of its orbit between the Scattered Disk and the inner Oort Cloud.



Sedna's closest approach to the Sun (perihelion) will be in 2076

Sedna will reach aphelion (farthest point from the Sun) in about 7776